

Through the Stable–Gaussian Endpoint

Uniform critical transseries, prefix-independent cutoff corrections, and conditional extremes

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Abstract

We study the critical countable-action equation $U(q) = \sum_{j \geq 1} w_j q^j e^{jU(q)}$, with nonnegative weights, $\sum jw_j = 1$, and an eventually exact tail $w_j = c_\varepsilon j^{-3+\varepsilon}$. The tail exponent approaches two while $|\varepsilon| \log n$ remains bounded. This is the joint-limit question explicitly left open in the companion marginal-critical manuscript. We prove a uniform, pole-cancelled inverse representation and a convergent correction chart about a deformed Lambert core. A direct Fourier argument gives the complete central coefficient through its first correction and a uniform local limit theorem. For action cutoffs we obtain a two-term formula in which all finite-prefix constants cancel. It yields a two-term minimal-budget formula and the exact criterion for asymptotically lossless truncation. Conditioned on the coefficient constraint, the rescaled large actions converge to a Poisson point process. The analysis works from both sides of the endpoint, including a triangular finite-variance regime in which the total variance is not the correct central normalization. Proofs are self-contained apart from explicitly identified classical analytic identities; reproducible algebraic and numerical checks are supplied.

Status and scope. The proposed contribution is the uniform joint-limit theory and its explicit correction formulas for the stated tail class, not the classical fixed-exponent stable or Gaussian laws. Publication priority has not been established. The article contains conventional proofs, not a new Lean formalization or an independently peer-reviewed result. The numerical checks are diagnostics, not interval certificates.

Contents

1	The research question and its precise resolution	2
1.1	The gap in the existing program	2
1.2	What is proved here	2
1.3	Classical results and the novelty boundary	3
2	The normalized family and exact coefficient identities	3
3	Cancelling the endpoint pole before taking limits	4
4	A uniform inverse and a convergent deformed-core chart	6
5	The moving fluctuation and extreme-action scales	7
6	A uniform local limit theorem with its first central correction	8
7	The uniform cutoff correction and cancellation of prefix data	10
8	A two-term minimal action budget	12
9	Conditional large actions converge to a Poisson process	13
10	Interpretation and the finite-variance-side surprise	14
10.1	Three different limits	14
10.2	Finite variance need not give the correct central scale	14
10.3	Why finite-prefix data cancel only in the displayed ratio	14
11	Reproducible checks and numerical diagnostics	15
11.1	What was actually checked	15
11.2	Large-order cutoff checks	15
11.3	Inverse checks	16
11.4	Reproduction and limitations	16
12	Proof dependencies, repository scope, and formalization	16
12.1	Dependency structure	16
12.2	Scope of the repository inspection	17
12.3	A realistic formalization route	17
13	Further research questions	17
14	Conclusion	18
A	A geometric truncation bound for the convergent chart	19
	References	19

1 The research question and its precise resolution

1.1 The gap in the existing program

The inspected PROVEIT transseries index describes a large canonical inversion volume, focused Lean modules, and separate research arrivals [2, 1]. Among those arrivals, *Beyond Finite-Action Folds* treats positive eventually exact tails $j^{-1-\alpha}$, for fixed $1 < \alpha < 2$, with stable coefficient laws and critical Hahn inversion [3]. The later companion *Marginal Critical Transseries* treats the endpoint $\alpha = 2$, including Lambert charts and cutoff corrections [4]. Its first proposed research question asks for joint inverse and coefficient expansions when $(2 - \alpha) \log n$ remains bounded, and specifically asks whether the prefix-independent correction has a uniform interpolating form. That is the target of this article.

Editorial note (ProveIt, 2026-09-29). The marginal companion is now filed in this series (batch 48) as `Marginal_Critical_Transseries_Action_Budgets/`. The same question is Question 1 of the logarithmic-endpoint package filed beside it [5], an independent treatment of the same endpoint, which this article did not see. The confluent package [6], filed with this one and written independently of it, answers the question at leading order for every rate $\varepsilon_n \rightarrow 0$ of either sign. The two agree term by term: its A_n and B_n are N and B here, its window forms are those of Section 10, its minimal budget is the leading term of Corollary 8.1 at $\alpha = 2$, its cutoff profile $e^{-1/(2s^2)}$ is the $\alpha \rightarrow 2$ limit of $\exp(-\ell^{-\alpha}/\alpha)$, and both prove the conditioned Poisson limit. Inside the window this article goes one order further. At $\varepsilon = 0$ its first cutoff correction is the marginal companion's critical correction and, after a change of logarithmic normalization, that of the logarithmic-endpoint package.

The difficulty is not solved by substituting $\alpha = 2$ in a stable formula. Two terms, one involving $\Gamma(-\alpha)$ and the other involving $\zeta(\alpha - 1)$, separately diverge. Their cancellation is of leading order in the joint limit. Keeping one and discarding the other changes the normalization. We keep their difference before either inversion or Fourier analysis.

1.2 What is proved here

Let $\varepsilon = 2 - \alpha$. Define the entire crossover function

$$\mathcal{H}(z) = \frac{e^z - 1}{z}, \quad \mathcal{H}(0) = 1, \quad F_\varepsilon(H) = H\mathcal{H}(\varepsilon H). \quad (1.1)$$

For coefficient order n , the main scales are determined by

$$B^2 = nc_\varepsilon F_\varepsilon(\log B), \quad H = \log B, \quad N = (nc_\varepsilon)^{1/(2-\varepsilon)}, \quad E = e^{\varepsilon H}, \quad F = F_\varepsilon(H), \quad r = E/F. \quad (1.2)$$

The root defining B is the large positive root. Under $|\varepsilon| \log n \leq K$, it exists uniquely in the range relevant below. Then B is the central fluctuation scale and N is the action-cutoff scale. They satisfy $N/B \asymp_K H^{-1/2}$.

Writing $u_n = [q^n]U$, and denoting by $u_{n,M}$ the coefficient after removing all actions larger than M without renormalizing the retained weights, we prove, uniformly for $\ell = M/N$ in compact subsets of $(0, \infty)$,

$$\frac{u_{n,M}}{u_n} = \exp\left(-\frac{\ell^{-\alpha}}{\alpha}\right) \left[1 + \frac{r}{4} \left\{\log \frac{2}{r\ell^2} + 1 - \gamma - \frac{2}{\ell^2}\right\} + O_K\left(\frac{(\log H)^2}{H^2}\right)\right]. \quad (1.3)$$

Here γ is Euler's constant; constants in the error may also depend on the chosen compact ℓ -interval and on the fixed weight family. No finite-prefix moment appears in the displayed correction. At $\varepsilon = 0$, $r = 1/H$, and this reduces to the correction in the marginal companion.

The result here is its uniform extension through the endpoint, with a proved remainder on both sides.

The inverse theorem, central coefficient theorem, cutoff theorem, minimal-budget corollary, and conditional point-process theorem are proved in Sections 4–9. None is deduced by an unjustified interchange of the limits in n and α .

1.3 Classical results and the novelty boundary

The Lagrange coefficient mechanism is classical. The connection between positive coefficient models, random allocations, and conditioned sums is also classical; Janson's treatment includes both fixed-index stable local limits and the $\sqrt{n \log n}$ endpoint law [8, Theorem 17.14 and Example 18.29]. The distinction between hard and soft truncation of heavy tails is an established subject [9]. We do not claim these mechanisms as new. The exact polylogarithm continuation formula is likewise classical [7, Eq. 25.12.12].

The statements offered as a model-specific extension are the combined *uniform* expansions, the first correction and its prefix cancellation, the corresponding convergent deformed-core chart, and the consequences for conditional actions in this moving-index regime. The literature inspection is targeted, not an exhaustive priority search. A proof in this article establishes its stated mathematical conclusion; it does not establish that no equivalent formula has appeared elsewhere.

2 The normalized family and exact coefficient identities

Definition 2.1 (Admissible finite-prefix family). Fix $\varepsilon_0 \in (0, 1/4)$ and an integer $J \geq 1$. For $|\varepsilon| < \varepsilon_0$, suppose

$$w_j(\varepsilon) = c_\varepsilon j^{-3+\varepsilon} + p_j(\varepsilon), \quad p_j = 0 \quad (j > J), \quad (2.1)$$

where $c_\varepsilon > 0$ and the finitely many p_j are real analytic near zero. Assume $w_j \geq 0$, w_1 is bounded away from zero, and

$$\sum_{j \geq 1} j w_j(\varepsilon) = 1. \quad (2.2)$$

All uniform assertions are made in a sufficiently small closed parameter interval on which these conditions hold. Individual p_j may be negative. Set

$$G_\varepsilon(z) = \sum_{j \geq 1} w_j z^j = c_\varepsilon \text{Li}_{3-\varepsilon}(z) + P_\varepsilon(z), \quad v_\varepsilon = \frac{1}{c_\varepsilon} \sum_{j=1}^J j^2 p_j, \quad (2.3)$$

where $P_\varepsilon(z) = \sum_{j=1}^J p_j z^j$.

A particularly transparent test family is

$$c_\varepsilon = \frac{1}{\zeta(2-\varepsilon) + \eta}, \quad p_1 = \eta c_\varepsilon, \quad p_j = 0 \quad (j > 1), \quad \eta \geq 0. \quad (2.4)$$

It has $v_\varepsilon = \eta$. Taking $\eta = 0$ gives the pure polylogarithm model; taking $\eta = 2$ changes only the raw first-action weight before critical normalization.

Let $U_\varepsilon \in q\mathbb{R}[[q]]$ be the solution of

$$U_\varepsilon(q) = G_\varepsilon(qe^{U_\varepsilon(q)}). \quad (2.5)$$

Coefficientwise substitution determines the solution uniquely, one degree at a time. All its coefficients are nonnegative. The cutoff solution uses $G_{\varepsilon,M}(z) = \sum_{j \leq M} w_j z^j$ in place of G_ε . The

parameter ε is fixed during each coefficient extraction, but may depend on n in an asymptotic statement.

Proposition 2.2 (Exact coefficient–Poisson correspondence). *Put $\rho_\varepsilon = \exp(-G_\varepsilon(1))$. For independent random variables $Z_j \sim \text{Poisson}(nw_j)$, define $S_n = \sum_{j \geq 1} jZ_j$ and $S_{n,M} = \sum_{j \leq M} jZ_j$. These sums are almost surely finite, and $\mathbb{E}S_n = n$. Then*

$$u_n = \frac{1}{n}[z^n]e^{nG_\varepsilon(z)} = \frac{\rho_\varepsilon^{-n}}{n}\mathbb{P}(S_n = n), \quad (2.6)$$

$$\frac{u_{n,M}}{u_n} = e^{-n\Lambda_M} \frac{\mathbb{P}(S_{n,M} = n)}{\mathbb{P}(S_n = n)}, \quad \Lambda_M = \sum_{j > M} w_j. \quad (2.7)$$

Equivalently, the ratio in (2.7) is $\mathbb{P}(Z_j = 0 \text{ for every } j > M \mid S_n = n)$.

Proof. Set $W = qe^U$; then $W = qe^{G(W)}$. Lagrange inversion gives

$$[q^n]G(W) = \frac{1}{n}[z^{n-1}]G'(z)e^{nG(z)} = \frac{1}{n^2}[z^{n-1}](e^{nG(z)})' = \frac{1}{n}[z^n]e^{nG(z)}.$$

The sum of the Poisson rates is finite, since $\sum w_j < \infty$. The total number of jumps is consequently finite almost surely. Its weighted mean is $n \sum jw_j = n$. The probability generating function of S_n is $\exp(n(G(z) - G(1)))$, proving the first identity. Apply the same argument to the cutoff and divide. Independence of the retained and deleted Poisson variables proves the conditional interpretation. $\square$

Proposition 2.3 (The real critical branch). *For $0 < q < \rho_\varepsilon$, the real branch is parameterized uniquely by*

$$W = e^{-s}, \quad q = e^{-s - G_\varepsilon(e^{-s})}, \quad U = G_\varepsilon(e^{-s}), \quad s > 0. \quad (2.8)$$

Its power series has radius ρ_ε . If $\Delta = \log(\rho_\varepsilon/q)$, then

$$\Delta = s + G_\varepsilon(e^{-s}) - G_\varepsilon(1), \quad U = G_\varepsilon(1) + \Delta - s. \quad (2.9)$$

Proof. The function $q(W) = We^{-G(W)}$ increases strictly on $(0, 1)$, since $WG'(W) < \sum jw_j = 1$. It runs from zero to ρ . The positive formal solution converges at every such q : its successive coefficientwise approximants are bounded by this real solution. Hence its radius is at least ρ . But $dU/dq = G'(W)e^{G(W)}/(1 - WG'(W))$ tends to infinity as $W \uparrow 1$, so the radius cannot exceed ρ . Taking logarithms gives (2.9). $\square$

In transseries notation, put $q = e^{-x}$. Then $U(e^{-x}) = \sum_{n \geq 1} u_n e^{-nx}$, convergent for $x > G_\varepsilon(1)$. The large-order questions concern these exponential-sector coefficients. The critical inverse concerns $\Delta = x - G_\varepsilon(1) \downarrow 0$, rather than a claim about arbitrary transseries or arbitrary summation procedures.

3 Cancelling the endpoint pole before taking limits

Define analytic functions of ε near zero by removable continuation:

$$g_\varepsilon = 2\Gamma(-2 + \varepsilon) - \frac{1}{\varepsilon}, \quad z_\varepsilon = \zeta(1 - \varepsilon) + \frac{1}{\varepsilon}, \quad a_\varepsilon = 1 + \varepsilon g_\varepsilon. \quad (3.1)$$

Their endpoint values are

$$g_0 = \frac{3}{2} - \gamma, \quad z_0 = \gamma, \quad a_0 = 1. \quad (3.2)$$

We write

$$K_\varepsilon(L) = F_\varepsilon(L) + g_\varepsilon e^{\varepsilon L} + z_\varepsilon + v_\varepsilon. \quad (3.3)$$

For $\varepsilon \neq 0$, this is exactly $2\Gamma(-2 + \varepsilon)e^{\varepsilon L} + \zeta(1 - \varepsilon) + v_\varepsilon$, but (3.3) is the stable expression at $\varepsilon = 0$. In particular,

$$K_0(L) = L + \frac{3}{2} + v_0, \quad K'_\varepsilon(L) = a_\varepsilon e^{\varepsilon L}. \quad (3.4)$$

Lemma 3.1 (Exact desingularized expansion). *There is a function $R_\varepsilon(s)$, jointly analytic near $(\varepsilon, s) = (0, 0)$, such that, with a consistent logarithm of s ,*

$$s + G_\varepsilon(e^{-s}) - G_\varepsilon(1) = \frac{c_\varepsilon s^2}{2} \{K_\varepsilon(\log(1/s)) + sR_\varepsilon(s)\}. \quad (3.5)$$

More explicitly,

$$R_\varepsilon(s) = \frac{2}{c_\varepsilon} \sum_{k \geq 3} \frac{(-1)^k}{k!} \left(c_\varepsilon \zeta(3 - \varepsilon - k) + \sum_{j=1}^J p_j j^k \right) s^{k-3}. \quad (3.6)$$

The series converges normally on sufficiently small product neighborhoods.

Proof. Use the classical convergent expansion, for $|s| < 2\pi$ on a logarithmic branch and nonintegral order,

$$\text{Li}_{3-\varepsilon}(e^{-s}) = \Gamma(-2 + \varepsilon) s^{2-\varepsilon} + \sum_{k \geq 0} \zeta(3 - \varepsilon - k) \frac{(-s)^k}{k!}.$$

This is the specialization of [7, Eq. 25.12.12]. The constant term cancels. The linear term cancels by (2.2). The quadratic and nonanalytic terms combine into $c_\varepsilon s^2 K_\varepsilon(\log(1/s))/2$. Terms of order at least three give (3.6). Their zeta factors have no parameter pole near zero; normal convergence on smaller disks follows from the same polylogarithm expansion, or from the zeta functional equation and the geometric factor $(|s|/(2\pi))^k$ times a polynomial in k . Thus the remaining parameter singularity is removable, proving the formula also at zero. Finite-prefix exponentials are entire in s . $\square$

Lemma 3.2 (Uniform logarithmic estimates). *Fix $K < \infty$. As $L \rightarrow \infty$, uniformly for $|\varepsilon|L \leq K$,*

$$F_\varepsilon(L) \asymp_K L, \quad e^{\varepsilon L} \asymp_K 1, \quad K_\varepsilon(L) = F_\varepsilon(L) + O_K(1), \quad (3.7)$$

$$K_\varepsilon(L) = F_\varepsilon(L) + \left(\frac{3}{2} - \gamma\right) e^{\varepsilon L} + \gamma + v_\varepsilon + O_K(L^{-1}). \quad (3.8)$$

For real $b = O(\log L)$,

$$F_\varepsilon(L + b) = F_\varepsilon(L) + e^{\varepsilon L} b + O_K\left(\frac{b^2}{L}\right). \quad (3.9)$$

Proof. The continuous positive function $\mathcal{H}$ has positive lower and finite upper bounds on $[-K, K]$. Analyticity gives $g_\varepsilon = g_0 + O(\varepsilon)$, $z_\varepsilon = \gamma + O(\varepsilon)$. Finally, $F'_\varepsilon = e^{\varepsilon L}$, $F''_\varepsilon = \varepsilon e^{\varepsilon L}$; Taylor's formula gives the last estimate. These arguments also give the analogous complex estimates with factors $e^{| \varepsilon b |}$ when needed below. $\square$

The term $(e^{\varepsilon L} - 1)/\varepsilon$, not $e^{\varepsilon L}/\varepsilon$, is the crossover kernel. The subtraction is indispensable when εL stays bounded. It retains the cancellation that is lost by taking a fixed-index stable approximation first.

4 A uniform inverse and a convergent deformed-core chart

Theorem 4.1 (Two-term joint inverse). *Let $\Delta \downarrow 0$, put $D = \log(1/\Delta)$, and assume $|\varepsilon|D \leq K$. Define*

$$F_* = F_\varepsilon(D/2), \quad E_* = e^{\varepsilon D/2}, \quad A_* = \frac{1}{2} \log \frac{c_\varepsilon F_*}{2}, \quad d_* = \left(\frac{3}{2} - \gamma\right)E_* + \gamma + v_\varepsilon. \quad (4.1)$$

The solution of (2.9) has the uniform expansion

$$s = \sqrt{\frac{2\Delta}{c_\varepsilon F_*}} \left[1 - \frac{E_* A_* + d_*}{2F_*} + O_K \left(\frac{(\log D)^2}{D^2} \right) \right]. \quad (4.2)$$

Consequently,

$$s = 2 \sqrt{\frac{\Delta}{c_\varepsilon D \mathcal{H}(\varepsilon D/2)}} \left[1 + O_K \left(\frac{\log D}{D} \right) \right]. \quad (4.3)$$

The corresponding solution of the original feedback equation is $U = G_\varepsilon(1) + \Delta - s$.

Proof. Write $L = \log(1/s)$. By (3.5),

$$2L = D + \log(c_\varepsilon/2) + \log\{K_\varepsilon(L) + e^{-L}R_\varepsilon(e^{-L})\}. \quad (4.4)$$

First bracket the solution with L equal to $D/2 + C \log D$, using Lemma 3.2 and two fixed constants C of opposite sufficiently large size. Monotonicity follows from Proposition 2.3; hence $L = D/2 + O_K(\log D)$. All the subsequent estimates are therefore within a fixed bounded εL -window.

Set $L = D/2 + A_* + b$. Initially the equation gives $b = O_K(\log D/D)$. By (3.9) and (3.8),

$$K_\varepsilon(L) = F_* + E_* A_* + d_* + O_K((\log D)^2/D).$$

The term $E_* b$ is within this error. The analytic remainder in (4.4) is exponentially smaller. Expanding the logarithm now yields

$$b = \frac{E_* A_* + d_*}{2F_*} + O_K((\log D)^2/D^2).$$

Exponentiate $s = e^{-D/2 - A_* - b}$. Since $e^{-D/2 - A_*} = \sqrt{2\Delta/(c_\varepsilon F_*)}$, the theorem follows. $\square$

The two-term formula is an asymptotic result. A different construction gives an actually convergent correction chart.

Theorem 4.2 (Uniform analytic correction about the deformed core). *For $|\varepsilon|D \leq K$ and small $\Delta > 0$, let L_0 be the unique large solution of*

$$\Delta = \frac{c_\varepsilon}{2} e^{-2L_0} K_\varepsilon(L_0), \quad s_0 = e^{-L_0}. \quad (4.5)$$

Put $X = s_0$, $Y = 1/K_\varepsilon(L_0)$, and $Z = e^{\varepsilon L_0}/K_\varepsilon(L_0)$. There is a single holomorphic germ $V(\varepsilon, X, Y, Z)$, on a product neighborhood independent of the particular joint-limit path, such that the exact solution is

$$s = s_0 \exp\{-V(\varepsilon, X, Y, Z)\}. \quad (4.6)$$

The germ has a convergent multivariate power series, is divisible by XY , and satisfies

$$V = \frac{XY R_\varepsilon(0)}{2 - a_\varepsilon Z} + O(X^2 Y) \quad (4.7)$$

in a smaller fixed polydisk. In particular, on the real joint-limit paths, $s/s_0 = 1 + O_K(s_0/\log(1/s_0))$.

Proof. Lemma 3.2 shows that the right side of (4.5) is positive and strictly decreasing at the large root: its logarithmic derivative is $-2 + K'_\varepsilon(L)/K_\varepsilon(L) = -2 + O_K(1/L)$. The same bracket used for Theorem 4.1 gives existence and uniqueness in the large-root region.

The exact identity

$$K_\varepsilon(L_0 + v) - K_\varepsilon(L_0) = a_\varepsilon e^{\varepsilon L_0} v \mathcal{H}(\varepsilon v)$$

turns (3.5), with $s = s_0 e^{-v}$, into

$$e^{-2v} [1 + a_\varepsilon Z v \mathcal{H}(\varepsilon v) + XY e^{-v} R_\varepsilon(X e^{-v})] = 1. \quad (4.8)$$

Its left side minus one is holomorphic in all five variables. At $(\varepsilon, X, Y, Z, v) = (0, 0, 0, 0, 0)$, the derivative with respect to v is -2 . The analytic implicit-function theorem gives one germ on a fixed polydisk, not a different germ for each asymptotic path. At $X = 0$ or $Y = 0$, its small solution is identically zero. Analytic division therefore gives the factor XY . Taking the coefficient of X in (4.8) gives (4.7); the remainder retains the factor Y . Finally, $X \rightarrow 0$, $Y, Z = O_K(1/L_0)$, so every sufficiently late real path lies inside that fixed polydisk. $\square$

Remark 4.3 (What is and is not a finite-generator statement). The correction V is a genuinely convergent finite-generator analytic chart. Its core is the inverse of (4.5); it is not asserted to be an elementary function, nor an ordinary Lambert function when $\varepsilon \neq 0$. We do not claim a global complex branch classification or a convergent expansion uniform over fixed nonzero ε as $D \rightarrow \infty$.

At $\varepsilon = 0$, put $d = 3/2 + v_0$. The core becomes $\Delta = (c_0/2)e^{-2L_0}(L_0 + d)$, so it is explicitly

$$s_0 = \left[\frac{-4\Delta}{c_0 W_{-1}(-4\Delta e^{-2d}/c_0)} \right]^{1/2}. \quad (4.9)$$

Thus Theorem 4.2 deforms the real negative Lambert branch continuously and analytically at the level of the desingularized equation. The identity follows directly by setting $y = 2(L_0 + d)$ and solving $ye^{-y} = 4\Delta e^{-2d}/c_0$.

5 The moving fluctuation and extreme-action scales

Lemma 5.1 (Existence, size, and exact scale identity). *For every fixed K , all sufficiently large n , and $|\varepsilon| \log n \leq K$, the equation $e^{2H} = nc_\varepsilon F_\varepsilon(H)$ has a unique root in $[\frac{2}{5} \log n, \frac{3}{5} \log n]$. At this root,*

$$H = \frac{1}{2} \log n + \frac{1}{2} \log \log n + O_K(1), \quad F \asymp_K H, \quad r \asymp_K H^{-1}. \quad (5.1)$$

With $\alpha = 2 - \varepsilon$,

$$\left(\frac{N}{B} \right)^\alpha = r, \quad \frac{N}{B} = r^{1/\alpha} \asymp_K H^{-1/2}. \quad (5.2)$$

Proof. On the displayed interval, $F_\varepsilon(H) \asymp_K \log n$. The function $2H - \log(nc_\varepsilon) - \log F_\varepsilon(H)$ changes sign at its endpoints. Its derivative is $2 - e^{\varepsilon H}/F_\varepsilon(H) > 1$ for large n . Taking logarithms of the defining equation gives (5.1). Finally,

$$(N/B)^\alpha = \frac{nc_\varepsilon}{B^{2-\varepsilon}} = \frac{B^\varepsilon}{F} = r.$$

Since $\varepsilon = O_K(1/H)$, $r^{1/\alpha}/\sqrt{r} = \exp(O_K(\log H/H))$, proving the comparison. $\square$

Corollary 5.2 (Explicit crossover function). *Write $\theta_n = \varepsilon \log n$. Uniformly for $|\theta_n| \leq K$,*

$$B^2 = \frac{c_\varepsilon n \log n}{2} \mathcal{H}(\theta_n/2) \left[1 + O_K \left(\frac{\log \log n}{\log n} \right) \right]. \quad (5.3)$$

If $\theta_n \rightarrow \theta$, then

$$\frac{N}{\sqrt{n}} \rightarrow \sqrt{c_0} e^{\theta/4}. \quad (5.4)$$

Proof. Insert $H = \frac{1}{2} \log n + O_K(\log \log n)$ into $F = H \mathcal{H}(\varepsilon H)$. The function $\mathcal{H}$ and its derivatives are bounded on the relevant compact interval. For the second formula, take logarithms:

$$\log(N/\sqrt{n}) = \frac{\log c_\varepsilon}{2 - \varepsilon} + \frac{\varepsilon \log n}{2(2 - \varepsilon)}.$$

Analyticity implies $c_\varepsilon \rightarrow c_0$. □

6 A uniform local limit theorem with its first central correction

We now prove the needed triangular-array limit directly. Applying a fixed-law central or stable limit theorem separately for each ε would not give the uniform result.

Lemma 6.1 (Uniform Fourier damping). *Let $\chi_n(t) = \mathbb{E} \exp(it(S_n - n)/B)$. For large n in the window, there exist positive constants b_1, b_2 , independent of n, ε , such that*

$$|\chi_n(t)| \leq e^{-b_1 t^2}, \quad |t| \leq B^{1/2}, \quad (6.1)$$

$$|\chi_n(t)| \leq e^{-b_2 n/B}, \quad B^{1/2} \leq |t| \leq \pi B. \quad (6.2)$$

The same estimates hold for $\mathbb{E} \exp(it(S_{n,M} - n)/B)$, uniformly when M/N is in a fixed compact subinterval of $(0, \infty)$.

Proof. The real logarithm of the modulus is $-n \sum w_j (1 - \cos(jt/B))$. On $|y| \leq 1$, $1 - \cos y \geq cy^2$. Retain actions $J < j \leq R$, where $R = \lfloor B/\max(1, |t|) \rfloor$, or the minimum of this number and M in the cutoff case. For $|t| \leq B^{1/2}$, this gives $\log R \geq H/2 - O(1)$, because $M \asymp N \gg B^{1/2}$. Integral comparison implies

$$\sum_{J < j \leq R} j^{-1+\varepsilon} \geq c_K F.$$

The retained exponent is therefore at most $-c_K n c_\varepsilon F t^2 / B^2 = -c_K t^2$. For the remaining interval, retain only $j = 1$. Since $|t/B| \leq \pi$, $1 - \cos(t/B) \geq ct^2/B^2$, and w_1 has a uniform positive lower bound. The exponent is at most $-cnt^2/B^2 \leq -cn/B$. □

Theorem 6.2 (Uniform local limit and central coefficient). *For the admissible family and $|\varepsilon| \log n \leq K$,*

$$\sup_{k \in \mathbb{Z}} |B \mathbb{P}(S_n = n + k) - \varphi(k/B)| \rightarrow 0, \quad \varphi(x) = \frac{e^{-x^2/2}}{\sqrt{2\pi}}, \quad (6.3)$$

uniformly in the parameter window. At the center the sharper expansion is

$$u_n = \frac{\rho_\varepsilon^{-n}}{\sqrt{2\pi} n B} \left[1 - \frac{1}{F} \left\{ \frac{E(1 - \gamma + \log 2)}{4} + \frac{\gamma + v_\varepsilon}{2} \right\} + O_K(H^{-2}) \right]. \quad (6.4)$$

Proof. For $t \neq 0$, use the principal logarithm of $-it$, and put $L_t = \log(-it)$. The exact cumulant is

$$\Psi_n(t) = n \sum_{j \geq 1} w_j (e^{itj/B} - 1 - itj/B).$$

Lemma 3.1, with $s = -it/B$, gives

$$\Psi_n(t) = -\frac{t^2}{2} + \frac{t^2}{2F} \{EL_t - (\tfrac{3}{2} - \gamma)E - \gamma - v_\varepsilon\} + \mathcal{E}_n(t). \quad (6.5)$$

Here we used

$$F_\varepsilon(H - L_t) = F - EL_t + O(|\varepsilon|E|L_t|^2 e^{|\varepsilon|L_t}).$$

On $0 < |t| \leq T = H^{1/12}$, the error is bounded by

$$|\mathcal{E}_n(t)| \leq \frac{C_K}{H^2} |t|^2 (1 + |\log |t||^2) \max(|t|^{1/4}, |t|^{-1/4}) + \frac{C_K |t|^3}{BF}. \quad (6.6)$$

The last term is the analytic remainder. The replacement of $g_\varepsilon, z_\varepsilon$ by their endpoint values contributes the first bound. We may shrink the parameter interval so that $|\varepsilon| \leq 1/4$. The singularity at $t = 0$ in this bound is integrable.

The displayed correction in (6.5) is uniformly small on this central interval. Expanding its exponential, and using Lemma 6.1, gives an integrated error $O_K(H^{-2})$. Indeed, its square is bounded by H^{-2} times an integrable Gaussian–power–logarithmic function. The integral of (6.6) has the same bound; $1/(BF)$ is smaller than every fixed negative power of H . The part $|t| > T$ contributes $O_K(e^{-bT^2}) + O(Be^{-bn/B})$, again smaller than H^{-2} . These estimates justify termwise integration, rather than merely a pointwise characteristic-function limit.

At the center, Fourier inversion is

$$B\mathbb{P}(S_n = n) = \frac{1}{2\pi} \int_{-\pi B}^{\pi B} e^{\Psi_n(t)} dt.$$

The imaginary part of L_t is odd and integrates to zero. For a standard normal random variable Z ,

$$\mathbb{E}(Z^2 \log |Z|) = 1 - \frac{\gamma}{2} - \frac{\log 2}{2}. \quad (6.7)$$

This follows by differentiating $\mathbb{E}|Z|^p = 2^{p/2} \Gamma((p+1)/2)/\sqrt{\pi}$ at $p = 2$. Consequently the normalized first correction is

$$\frac{1}{2F} \left[E \left(1 - \frac{\gamma}{2} - \frac{\log 2}{2} \right) - (\tfrac{3}{2} - \gamma)E - \gamma - v_\varepsilon \right] = -\frac{1}{F} \left[\frac{E(1 - \gamma + \log 2)}{4} + \frac{\gamma + v_\varepsilon}{2} \right].$$

Combine this with (2.6).

Finally, the same bounds prove that the characteristic function, extended by zero outside $[-\pi B, \pi B]$, converges to $e^{-t^2/2}$ in $L^1(\mathbb{R})$, uniformly in the window. Fourier inversion at arbitrary k/B then gives (6.3), since the exponential Fourier factor always has modulus one. $\square$

Corollary 6.3 (Explicit leading coefficient). *With $\theta_n = \varepsilon \log n$,*

$$u_n = \frac{\rho_\varepsilon^{-n}}{n^{3/2} \sqrt{\pi c_\varepsilon \log n \mathcal{H}(\theta_n/2)}} \left[1 + O_K \left(\frac{\log \log n}{\log n} \right) \right]. \quad (6.8)$$

The exponential factor ρ_ε^{-n} must be retained at the actual parameter. Replacing it by ρ_0^{-n} is generally invalid even when $\varepsilon = O(1/\log n)$, because an $O(\varepsilon)$ change in $\log \rho$ is multiplied by n .

7 The uniform cutoff correction and cancellation of prefix data

Lemma 7.1 (Deleted mean and retained moments). *Assume $\ell = M/N \in [\ell_-, \ell_+] \subset (0, \infty)$, and set*

$$b = \log(M/B) = \log \ell + \alpha^{-1} \log r, \quad d_M = \frac{n - \mathbb{E} S_{n,M}}{B}, \quad V_M = \frac{\text{Var } S_{n,M}}{B^2}. \quad (7.1)$$

Uniformly in this range,

$$n\Lambda_M = \ell^{-\alpha}/\alpha + O_K(M^{-1}), \quad (7.2)$$

$$d_M = \frac{r^{1/\alpha} \ell^{1-\alpha}}{\alpha - 1} + O_K(B^{-1}), \quad d_M^2 = \frac{r}{\ell^2} + O_K(\log H/H^2), \quad (7.3)$$

$$V_M = 1 + \frac{Eb + \gamma + v_\varepsilon}{F} + O_K((\log H)^2/H^2). \quad (7.4)$$

For $k = 3, 4$, the standardized retained cumulants satisfy

$$\kappa_{k,M} := \frac{n}{B^k} \sum_{j \leq M} j^k w_j = O_K(H^{-k/2}). \quad (7.5)$$

Proof. For large n , $M > J$, so the deleted tail is exactly the power tail. Integral comparison gives, uniformly near $\alpha = 2$,

$$\sum_{j > M} j^{-1-\alpha} = M^{-\alpha}/\alpha + O(M^{-1-\alpha}), \quad \sum_{j > M} j^{-\alpha} = M^{1-\alpha}/(\alpha - 1) + O(M^{-\alpha}).$$

Use $nc_\varepsilon = N^\alpha$ and (5.2). For the squared deleted mean, the replacements

$$r^{2/\alpha} = r\{1 + O_K(\log H/H)\}, \quad \ell^{2(1-\alpha)} = \ell^{-2}\{1 + O_K(H^{-1})\}, \quad (\alpha - 1)^{-2} = 1 + O_K(H^{-1})$$

give the stated error.

Uniform integral comparison with one endpoint correction gives

$$\sum_{j \leq M} j^{-1+\varepsilon} = F_\varepsilon(\log M) + \gamma + O(|\varepsilon|) + O(M^{-1+\varepsilon}). \quad (7.6)$$

For completeness, subtract the integral of $x^{-1+\varepsilon}$ on $[1, M]$. The resulting sum of successive sum-integral errors has terms $O(j^{-2+\varepsilon})$, and its parameter derivative has terms $O(j^{-2+\varepsilon} \log j)$. Both are uniformly summable near zero. The limiting constant is therefore $\gamma + O(|\varepsilon|)$; the remaining endpoint error is $O(M^{-1+\varepsilon})$. This proves (7.6) without any interchange of divergent sums.

Add the finite-prefix second moment $c_\varepsilon v_\varepsilon$, divide by $c_\varepsilon F$, and apply (3.9) with $b = O_K(\log H)$. This proves (7.4). Finally,

$$\frac{nc_\varepsilon M^{k-\alpha}}{B^k} = \ell^{k-\alpha} r^{k/\alpha} = O_K(H^{-k/2}), \quad k = 3, 4.$$

The finite-prefix contribution $O(n/B^k)$ is smaller. Integral comparison for the retained powers completes the proof. $\square$

Lemma 7.2 (Central probability for the cutoff). *In the same parameter range,*

$$B\sqrt{2\pi} \mathbb{P}(S_{n,M} = n) = 1 - \frac{Eb + \gamma + v_\varepsilon}{2F} - \frac{E}{2F\ell^2} + O_K((\log H)^2/H^2). \quad (7.7)$$

Proof. The real and imaginary parts of the centered cutoff cumulant obey

$$\begin{aligned}\operatorname{Re} \Psi_{n,M}(t) &= -V_M t^2/2 + O(\kappa_{4,M} t^4), \\ \Im \Psi_{n,M}(t) &= -d_M t + O(\kappa_{3,M} |t|^3).\end{aligned}$$

These inequalities follow directly from the Taylor remainders for cosine and sine. Fourier inversion at the center integrates $e^{\operatorname{Re} \Psi_{n,M}} \cos(\Im \Psi_{n,M})$. On $|t| \leq H^{1/12}$, put $a_M = V_M - 1$. Its expansion through the required order is

$$e^{-t^2/2} \{1 - a_M t^2/2 - d_M^2 t^2/2\}.$$

The integrated error is bounded by a fixed constant times

$$a_M^2 + \kappa_{4,M} + d_M^4 + |d_M| \kappa_{3,M} + \kappa_{3,M}^2 = O_K((\log H)^2/H^2).$$

For the phase estimate, use $|\cos(x+y) - \cos x| \leq |xy| + y^2/2$. This is why an apparent cubic error of size $H^{-3/2}$ does not survive at the center: it is multiplied by the deleted-mean shift or enters quadratically. The tail of the Fourier integral is negligible by Lemma 6.1. Integrating $t^2 e^{-t^2/2}$, then using Lemma 7.1, proves the result. $\square$

Theorem 7.3 (Uniform prefix-independent cutoff correction). *For every admissible family, uniformly for $|\varepsilon| \log n \leq K$ and $\ell = M/N \in [\ell_-, \ell_+] \subset (0, \infty)$,*

$$\frac{u_{n,M}}{u_n} = e^{-\ell^{-\alpha}/\alpha} \left[1 + \frac{r}{4} \left\{ \log \frac{2}{r\ell^2} + 1 - \gamma - \frac{2}{\ell^2} \right\} + O_K \left(\frac{(\log H)^2}{H^2} \right) \right]. \quad (7.8)$$

In particular, the displayed normalized correction is the same for all finite-prefix changes, after using that family's actual c_ε , N , and B .

Proof. Write the full central probability as $(B\sqrt{2\pi})^{-1}(1 + C/F + O(H^{-2}))$, where

$$C = -\frac{E(1 - \gamma + \log 2)}{4} - \frac{\gamma + v_\varepsilon}{2}.$$

Divide (7.7) by this expression. The numerator's $-(\gamma + v_\varepsilon)/(2F)$ cancels the identical denominator term. What remains is

$$\frac{\mathbb{P}(S_{n,M} = n)}{\mathbb{P}(S_n = n)} = 1 + \frac{E}{F} \left\{ -\frac{b}{2} + \frac{1 - \gamma + \log 2}{4} - \frac{1}{2\ell^2} \right\} + O_K((\log H)^2/H^2). \quad (7.9)$$

Now $b = \log \ell + \alpha^{-1} \log r$, and $\alpha^{-1} = 1/2 + O_K(H^{-1})$. Multiplication by $r = O_K(H^{-1})$ shows that replacing $\alpha^{-1} \log r$ by $\frac{1}{2} \log r$ costs only $O_K(\log H/H^2)$. Therefore the displayed term in (7.9) becomes

$$\frac{r}{4} \left\{ \log \frac{2}{r\ell^2} + 1 - \gamma - \frac{2}{\ell^2} \right\}.$$

Lastly, (7.2) replaces the survival factor in (2.7) by $e^{-\ell^{-\alpha}/\alpha}$ with relative error $O_K(M^{-1})$, smaller than the stated remainder. $\square$

Remark 7.4 (The exponent in the survival factor must not be frozen). Although $\alpha \rightarrow 2$, replacing $e^{-\ell^{-\alpha}/\alpha}$ by $e^{-1/(2\ell^2)}$ can change a term of order $1/H$. Such a replacement is allowed for the limiting law, but not generally for the first correction. Formula (7.8) keeps this contribution exactly.

Corollary 7.5 (Lossless truncation and the raw square-root scale). *In the bounded window,*

$$\frac{u_{n,M_n}}{u_n} \longrightarrow 1 \iff \frac{M_n}{N_n} \longrightarrow \infty. \quad (7.10)$$

If $M_n/N_n \rightarrow 0$, the ratio tends to zero. If $M_n/N_n \rightarrow \ell > 0$, the limit is $e^{-1/(2\ell^2)}$. Equivalently, when $\varepsilon \log n \rightarrow \theta$ and $M_n/\sqrt{n} \rightarrow m > 0$,

$$\frac{u_{n,M_n}}{u_n} \longrightarrow \exp\left(-\frac{c_0 e^{\theta/2}}{2m^2}\right). \quad (7.11)$$

Proof. The coefficient ratio is increasing in M and lies in $[0, 1]$. For each fixed $L > 0$, Theorem 7.3 applies at $M = \lfloor LN \rfloor$. If $M_n/N_n \rightarrow \infty$, use this cutoff as a lower bound and then let $L \rightarrow \infty$. Conversely, a bounded subsequence of M_n/N_n is bounded above by some finite L , giving an upper limit at most $e^{-1/(2L^2)} < 1$. The zero-limit statement follows by an upper bound at each fixed small L and then $L \downarrow 0$. The remaining assertions follow directly from the theorem and (5.4). $\square$

8 A two-term minimal action budget

Fix a target retained fraction $\tau \in (0, 1)$, independent of n . Since actions above n cannot contribute to $[q^n]$, there is always an integer $M \leq n$ retaining the whole coefficient. Thus define

$$M_{\min}(n, \varepsilon; \tau) = \min\{M \geq 1 : u_{n,M}/u_n \geq \tau\}. \quad (8.1)$$

Corollary 8.1 (Uniform two-term budget). *Let $L_\tau = -\log \tau$ and $\ell_0 = (\alpha L_\tau)^{-1/\alpha}$. In the bounded parameter window,*

$$\begin{aligned} \frac{M_{\min}}{N} = \ell_0 \left[1 - \frac{r}{4\alpha L_\tau} \left\{ \log \frac{2}{r\ell_0^2} + 1 - \gamma - \frac{2}{\ell_0^2} \right\} \right. \\ \left. + O_{K,\tau} \left(\frac{(\log H)^2}{H^2} \right) \right]. \end{aligned} \quad (8.2)$$

Proof. The leading cutoff law brackets $M_{\min}/N$ inside a fixed compact interval around ℓ_0 . The logarithm of (7.8) is

$$-\ell^{-\alpha}/\alpha + \frac{r}{4}h(\ell, r) + O_K((\log H)^2/H^2), \quad h(\ell, r) = \log \frac{2}{r\ell^2} + 1 - \gamma - \frac{2}{\ell^2}.$$

The derivative of the leading term at ℓ_0 is $\ell_0^{-\alpha-1} > 0$. The first displacement of its crossing of $\log \tau$ is therefore $-r\ell_0^{\alpha+1}h(\ell_0, r)/4$. Taylor's formula on that compact interval bounds the remaining displacement by $O_{K,\tau}((\log H)^2/H^2)$. Monotonicity brackets the integer crossing between the two continuous error bounds; the rounding error is at most $1/N$, which is smaller. Since $\ell_0^\alpha = (\alpha L_\tau)^{-1}$, this is (8.2). $\square$

This is an asymptotic budget, not a finite- n interval certificate. For an actual finite computation, a positive coefficient recurrence gives a separate way of testing the requested retained fraction. Section 11 describes such a recurrence. A uniform numerical value for the constants in the asymptotic error has not been certified here.

9 Conditional large actions converge to a Poisson process

The cutoff theorem is a largest-action statement. The local limit theorem allows an extension to all macroscopic extreme actions at once.

Theorem 9.1 (Conditional Poisson limit). *Let $\varepsilon = \varepsilon_n$ with $|\varepsilon_n| \log n \leq K$. Conditional on $S_n = n$, define the random counting measure*

$$\Xi_n = \sum_{j \geq 1} Z_j \delta_{j/N} \quad \text{on } (0, \infty). \quad (9.1)$$

Then Ξ_n converges in distribution, for the vague topology on locally finite counting measures, to a Poisson point process with intensity

$$\nu(dx) = x^{-3} dx. \quad (9.2)$$

In particular, if $J_{(k)}$ is the k -th largest action, counting multiplicities and padded with zeros, then for each $\ell > 0$,

$$\mathbb{P}(J_{(k)}/N \leq \ell \mid S_n = n) \longrightarrow e^{-1/(2\ell^2)} \sum_{m=0}^{k-1} \frac{(1/(2\ell^2))^m}{m!}. \quad (9.3)$$

Proof. For finitely many distinct indices and integers $m_j \geq 0$, the Poisson factorial-moment identity is exact:

$$\mathbb{E} \left[\prod_j (Z_j)_{m_j} \mid S_n = n \right] = \prod_j (nw_j)^{m_j} \frac{\mathbb{P}(S_n = n - \sum_j j m_j)}{\mathbb{P}(S_n = n)}. \quad (9.4)$$

It follows by shifting each count in its Poisson probability mass function; $(z)_m = z(z-1)\cdots(z-m+1)$.

Consider disjoint intervals $I_1, \dots, I_d$ whose closures lie in $(0, \infty)$. A fixed mixed factorial moment of their counts has shifts of size $O(N)$. Since $N/B \rightarrow 0$, Theorem 6.2 gives

$$\frac{\mathbb{P}(S_n = n - k)}{\mathbb{P}(S_n = n)} \longrightarrow 1 \quad \text{uniformly for } |k| \leq CN$$

for every fixed C . Summing (9.4) over the indices in the intervals yields the mixed factorial moments of independent Poisson variables, because

$$n \sum_{j/N \in I_a} w_j = \frac{1}{N} \sum_{j/N \in I_a} (j/N)^{-1-\alpha} \longrightarrow \int_{I_a} x^{-3} dx. \quad (9.5)$$

The finite prefix is eventually outside every such interval.

There is no moment-determinacy gap: the uniform local limit also bounds $\sup_k \mathbb{P}(S_n = k)/\mathbb{P}(S_n = n)$ by a fixed constant. Thus every mixed factorial moment is bounded by that constant times the corresponding product of the unconditioned means to the required powers. The exponential series for the joint count generating function is consequently dominated on a neighborhood of one. Its limit is the independent-Poisson generating function. The same first-moment bound gives tightness of the number of points in each compact set. Approximation of a nonnegative continuous compactly supported test function by interval step functions proves convergence of the Laplace functionals, and hence the asserted vague limit.

To pass from compact intervals to the entire tail (ℓ, ∞) , note from the same identity and probability-ratio bound that

$$\mathbb{E}[\Xi_n((L, \infty)) \mid S_n = n] \leq Cn\Lambda_{[LN]} \leq C_K L^{-7/4}$$

for $L \geq 1$ and all sufficiently large n . The omitted upper tail is therefore negligible as $L \rightarrow \infty$. The limiting number of actions above ℓ is Poisson with mean $\int_{\ell}^{\infty} x^{-3} dx = 1/(2\ell^2)$. At most $k-1$ such actions is exactly the event in (9.3). $\square$

The conditioning is essential in the formulation: coefficients are not unconditioned probabilities. The proof shows why it becomes invisible at the extreme-action scale in this window. Any fixed number of extreme actions shifts the total by $O(N) = o(B)$, too little to alter the leading local Gaussian density. The first cutoff correction measures part of the remaining, smaller interaction.

10 Interpretation and the finite-variance-side surprise

10.1 Three different limits

For fixed $1 < \alpha < 2$, the classical coefficient mechanism is stable, with a power scale $n^{1/\alpha}$. At exactly $\alpha = 2$, the central scale is $\sqrt{n \log n}$. In the window of this article, $\alpha = 2 - \theta_n / \log n$, the central limit is Gaussian but its scale retains the nontrivial factor $\mathcal{H}(\theta_n/2)$. These are three different asymptotic assertions. A bounded-window theorem does not by itself justify letting $\theta_n \rightarrow +\infty$ and claiming recovery of a fixed-index stable law.

The action scale is different again: $N \sim \sqrt{c_0 n} e^{\theta/4}$, whereas $B^2 \sim (c_0 n \log n/2) \mathcal{H}(\theta/2)$. Thus a cutoff much larger than $\sqrt{n}$, in the normalized sense of Corollary 7.5, preserves the coefficient even when it is much smaller than the central fluctuation scale. Confusing a largest-jump scale with a sum-fluctuation scale overestimates the required action budget.

10.2 Finite variance need not give the correct central scale

Take $\varepsilon_n = -\kappa / \log n$, with fixed $\kappa > 0$. Every member of this triangular family has $\alpha_n > 2$ and finite variance. Nevertheless, its total variance is not asymptotic to B^2 . Indeed,

$$\text{Var } S_n = n c_{\varepsilon} \{ \zeta(1 - \varepsilon) + v_{\varepsilon} \}, \quad \frac{\text{Var } S_n}{B^2} \longrightarrow \frac{1}{1 - e^{-\kappa/2}}. \quad (10.1)$$

This follows from the zeta pole and (5.3). Theorem 6.2, together with its characteristic-function proof, implies $(S_n - n)/B \Rightarrow N(0, 1)$. Normalizing instead by the actual standard deviation therefore gives a Gaussian with limiting variance $1 - e^{-\kappa/2}$, not one.

There is no contradiction. Rare actions far beyond the central scale retain a nonvanishing share of the second moment, although their probabilities vanish. For example, for fixed $a > 0$, integral comparison gives

$$\frac{n c_{\varepsilon} \sum_{j > aB} j^{-1+\varepsilon}}{\text{Var } S_n} \longrightarrow e^{-\kappa/2}.$$

The numerator is the contribution of those jumps to the Poisson variance. This is an explicit failure of uniform integrability of the normalized second moments. A fixed-distribution finite-variance intuition is therefore not an adequate substitute for the triangular Fourier proof.

10.3 Why finite-prefix data cancel only in the displayed ratio

The inverse expansion contains v_{ε} , and the central coefficient contains v_{ε} . Neither is prefix-independent. The ratio is special: the same retained finite-prefix variance enters its numerator and denominator with the same coefficient, and disappears upon division. Prefix changes also alter c_{ε} , ρ_{ε} , B , and N . The theorem never claims invariance when these normalizations are held artificially fixed. Nor does it assert that prefix data cancel to every higher order.

11 Reproducible checks and numerical diagnostics

11.1 What was actually checked

The supplied program `code/verify.py` executed 17 exact symbolic checks: 14 comparisons between direct formal iteration of the feedback equation and the Lagrange coefficient identity, plus three checks of the Gaussian constant and the prefix-cancellation algebra. The formal comparisons use two rational finite-action families and degrees one through seven. They test finite identities; they do not certify the analytic theorems.

For an independent finite- n diagnostic, if $p_k = \mathbb{P}(S_n = k)$, differentiation of the probability generating function gives the positive recurrence

$$p_0 = e^{-nG(1)}, \quad p_k = \frac{n}{k} \sum_{j=1}^k j w_j p_{k-j}. \quad (11.1)$$

The cutoff version restricts the sum to $j \leq M$ and changes p_0 accordingly. The program ran 18 such diagnostics, for $n = 512, 2048, 4096$, $\theta = \varepsilon \log n \in \{-1, 0, 1\}$, and $\eta \in \{0, 2\}$ in (2.4). In six cross-check cases at $n = 512$, recurrence and independently evaluated central Fourier integrals agreed with maximum absolute discrepancy $3.3306690738754696 \times 10^{-16}$, across the tested ratios and normalized central probabilities.

11.2 Large-order cutoff checks

The larger orders in Table 1 are evaluated by a scaled Fourier integral, not by constructing 10^{32} coefficients. The full cumulant uses the desingularized polylogarithm formula and 16 analytic terms. The cutoff cumulant uses 60 terms, with finite moments evaluated by direct sums or 14-term Euler–Maclaurin evaluation. Numerical integration is over $0 \leq t \leq 14$. These implementation truncations and floating-point quadrature estimates are not rigorous enclosures.

All displayed cutoff runs take M to be the nearest integer to N , but evaluate both approximations at the actual $\ell = M/N$. The column “leading” keeps α in the survival exponential. “Corrected” uses the full displayed expression in Theorem 7.3.

n	θ	Computed ratio	Leading	Corrected
10^8	-2	0.639884263	0.622371360	0.639837832
10^8	0	0.628290097	0.606533148	0.627967224
10^8	2	0.610090576	0.589362780	0.611719991
10^{16}	-2	0.626485822	0.614597962	0.626437090
10^{16}	0	0.622905481	0.606530659	0.622684654
10^{16}	2	0.618154572	0.598128083	0.617962252
10^{32}	-2	0.618213163	0.610604999	0.618168907
10^{32}	0	0.617565238	0.606530660	0.617441970
10^{32}	2	0.617011900	0.602372524	0.616782127

Table 1: Uniform-window cutoff diagnostics for the pure family. These are floating-point evaluations, not certified exact values.

The convergence is logarithmically slow. A very large value of n need not make the uncorrected cutoff law highly accurate. For example, at $n = 10^{16}$, $\theta = 2$, the computed ratio is approximately 0.618154572, compared with 0.598128083 from the leading term and 0.617962252 from the correction. This is evidence consistent with the formula, not a substitute for its uniform error proof.

11.3 Inverse checks

For the inverse diagnostics, $D = \log(1/\Delta)$ and $\theta = \varepsilon D$. A high-precision root solve in the variable $L = \log(1/s)$, using the desingularized expression, was compared with both inverse approximations. The analytic remainder was evaluated through order 16. Table 2 gives signed relative errors, approximation divided by the computed reference value minus one.

D	θ	Leading relative error	Corrected relative error
20	-2	0.07631914	-0.01277858
20	0	0.10242388	-0.01089807
20	2	0.17846262	-0.00151471
40	-2	0.04410638	-0.00418078
40	0	0.05972766	-0.00392070
40	2	0.09761422	-0.00194281
80	-2	0.02490134	-0.00133200
80	0	0.03419827	-0.00133957
80	2	0.05423873	-0.00095531

Table 2: Two-term inverse diagnostics for $\eta = 0$. The joint parameter here is $\theta = \varepsilon D$, not $\varepsilon \log n$.

11.4 Reproduction and limitations

The recorded environment was Python 3.13.5, NumPy 2.3.5, mpmath 1.3.0, SciPy 1.17.0, and SymPy 1.14.0. Constants and finite moments were evaluated with 75 mpmath decimal digits; the recurrence used NumPy long double; quadrature used ordinary double precision. The complete CSV data, exact-check list, environment record, and run summary are included. The default recurrence order is deliberately limited to avoid underflow of its initial Poisson mass on the recorded platform. The script reports underflow rather than silently returning a spurious answer.

The numerical program is not a Lean proof, a certified arbitrary-precision interval implementation, or a proof of the asymptotic error constants. Theorems 4.1, 4.2, 6.2, 7.3, and 9.1 stand on their written proofs.

12 Proof dependencies, repository scope, and formalization

12.1 Dependency structure

The inverse theory depends on the exact polylogarithm expansion, critical normalization, elementary uniform estimates, and the analytic implicit-function theorem. The coefficient theory depends on the Lagrange identity, the same desingularized expansion, Fourier inversion, and the explicit damping lemma. The cutoff correction then uses only retained moments and division of the two local expansions. The minimal budget uses monotonicity. The point-process result uses the uniform local limit and an exact Poisson factorial-moment identity.

No result from the companion marginal manuscript is used as a proof premise. Its open question is the research target, and its displayed endpoint formula is an overlap check. The argument here rederives that endpoint while proving uniformity in the moving parameter. No unverified theorem from the full canonical PROVEIT volume is needed.

12.2 Scope of the repository inspection

The inspected public snapshot is commit `0c973d8f5e7ef4550f4df1bf486d890037fd9f14`. The inspection covered the transseries project README, the opening portion of the group index, the relevant package inventory, and the complete README for the fixed-stable critical-Hahn package. The latter identifies its theorem scope and proof boundaries. The private companion `Marginal_Critical_Transseries.tex` was inspected through its abstract, main-result summary, and explicit further-research question. We did not audit the entire canonical volume or all source files in the repository. No repository files were modified.

This separation matters. An entry in the Lean crosswalk is evidence about that entry, not automatic formal verification of a neighboring analytic research claim. The present results are not represented as already available in the repository's Lean development.

12.3 A realistic formalization route

A useful Lean development can be split into four interfaces. First, formalize the finite coefficient identity and positivity using formal power series and a truncated exponential. Second, formalize the elementary scale algebra, including $(N/B)^\alpha = r$, and the cancellation of the finite-prefix moment. Third, supply analytic interfaces for the removable polylogarithm kernel and the parameter-uniform implicit theorem. Fourth, formalize a reusable lattice Fourier lemma with integrable remainder and damping hypotheses, and instantiate it for the full and cutoff sums.

The hard analytic obligations should remain explicit: normal convergence in the parameter, a uniform Fourier tail, the integrated central remainder, and the probability-ratio bound used in conditional factorial moments. Verifying a symbolic coefficient identity does not discharge any of these. The supplied exact checks suggest small initial formalization targets, not a claim of an existing machine-checked asymptotic proof.

13 Further research questions

1. Leave the bounded window and match the stable law. Extend the argument to $\varepsilon \log n \rightarrow +\infty$ while $\varepsilon \rightarrow 0$, with explicit uniform errors. Determine the maximal range in which a Gaussian approximation remains accurate after the deformed normalization, and construct an overlap expansion with the fixed-index stable density. The constants hidden in the bounded-window estimates cannot simply be assumed uniform as the window grows.

2. Compute the next correction and locate prefix reentry. Expand the full Fourier cumulant to order H^{-2} , retaining the parameter variation of the regularized gamma and zeta constants. Combine it with the fourth retained cumulant and the squared deleted-mean shift. Determine exactly which finite-prefix moments survive at that order. The cancellation proved here gives a nontrivial check on any such calculation.

3. Add a slowly varying tail. Replace the eventually exact tail by $cj^{-3+\varepsilon}L(j)$, with quantified uniform regularity of L . Identify necessary and sufficient conditions for the ratio between the extreme-action and central scales to vanish. Establish which slowly varying perturbations preserve a finite-generator inverse chart and which require new core inverses. A pointwise tail asymptotic alone is unlikely to control the first correction.

4. Allow a drifting distance from criticality. Replace $\sum jw_j = 1$ by a mean differing from one on the B/n scale. The leading local Gaussian is then sampled away from zero. Determine a joint coupling–index–cutoff expansion, including whether the odd Fourier term creates a prefix-dependent correction before order H^{-2} . The cancellation of the cubic contribution at the center must be revisited.

5. Quantify conditional Poisson convergence. The point-process theorem is qualitative. Derive explicit total-variation or bounded-test-function rates for counts on intervals separated from zero, and compute the first conditioning-induced covariance between two disjoint large-action windows. The exact factorial-moment formula and the shifted local Fourier expansion provide a direct route.

6. Develop complex continuation of the deformed core. The correction germ is holomorphic, but the core is selected here on its large real branch. Classify its nearby complex branches on a logarithmic cover, locate parameter-dependent turning points, and compare the monodromy with the Lambert endpoint and nonintegral Hahn regimes. A global two-sheet or Stokes statement would require additional work; it is not implicit in Theorem 4.2.

7. Replace asymptotic budgets by certified finite budgets. Make every constant in the damping and Taylor estimates explicit, and combine them with interval arithmetic for the positive recurrence or Fourier integrals. The goal is a proof-producing routine returning a guaranteed smallest or near-smallest M for a prescribed coefficient fraction. The current two-term formula is a principled initial estimate, not that certificate.

8. Signed and multi-type feedback. For signed weights the Poisson model and coefficient monotonicity disappear; for multiple feedback variables the criticality condition becomes a spectral condition. Determine which pole-cancellation and inverse-chart arguments survive, and identify replacement hypotheses for the scalar positive Fourier proof. These are distinct extensions, not consequences of analytic continuation of the positive-weight formulas.

14 Conclusion

The joint stable–Gaussian endpoint question has an explicit answer for positive finite-prefix power-tail feedback. The correct shared kernel is $(e^{\varepsilon L} - 1)/\varepsilon$. It simultaneously controls critical inversion and central coefficient growth. The associated extreme-action scale is smaller than the fluctuation scale by a logarithmic factor. At that smaller scale, the first cutoff correction is universal after normalization: finite-prefix second moments cancel, even though they remain visible in the two separate local expansions. The same separation of scales makes the conditioned large actions asymptotically Poisson.

These conclusions resolve the specified bounded-window follow-up, including its first interpolating cutoff correction, within the stated family and with conventional proofs. They do not assert an unrestricted transseries theorem, a global priority result, or a formal verification that has not been carried out. The exact formulas and explicit proof obligations make the next extensions concrete.

A A geometric truncation bound for the convergent chart

The analyticity statement in Theorem 4.2 implies an ordinary geometric error bound, distinct from the large-order asymptotic errors. After shrinking its domain, there exist constants $R_0, C > 0$ such that, uniformly for $|\varepsilon| \leq \delta$,

$$V(\varepsilon, X, Y, Z) = \sum_{a,b,c \geq 0} v_{abc}(\varepsilon) X^a Y^b Z^c, \quad |v_{abc}(\varepsilon)| \leq C R_0^{-a-b-c}.$$

Let $V_{\leq m}$ be its truncation to total degree at most m , and put $t = \max(|X|, |Y|, |Z|)/R_0 < 1$. Cauchy's estimate gives

$$|V - V_{\leq m}| \leq C \sum_{k > m} \binom{k+2}{2} t^k \leq C \frac{(m+3)^2 t^{m+1}}{(1-t)^3}. \quad (\text{A.1})$$

The second inequality follows by writing $k = m+1+l$, bounding $\binom{m+l+3}{2}$ by $(m+3)^2(l+1)^2/2$, and summing $\sum_{l \geq 0} (l+1)^2 t^l = (1+t)/(1-t)^3$. A bound on the relative error of $s_0 e^{-V_{\leq m}}$ follows from $|e^z - 1| \leq |z|e^{|z|}$. This is a convergent analytic truncation statement. It does not turn the H^{-1} coefficient expansion into a convergent series, nor supply numerical values of C, R_0 without a quantitative implicit-function analysis.

References

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- [3] *Beyond Finite-Action Folds: Critical Hahn Transseries, Stable Sector Asymptotics, and Sharp Action Budgets*, research package prepared for Vladimir Reshetnikov, 29 September 2026. Package README inspected at the snapshot above; used for scope comparison, not as a proof premise. https://github.com/VladimirReshetnikov/ProveIt/tree/0c973d8f5e7ef4550f4df1bf486d890037fd9f14/Analysis/Transseries/docs/series-and-transseries/Critical_Hahn_Transseries_Beyond_Finite_Action_Folds.
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- [5] [Editorial addition, ProveIt, 2026-09-29.] *The Logarithmic Critical Endpoint: Convergent Lambert Charts, All-Order Sector Laws, and Sharp Action-Cutoff Corrections*, research package filed in batch 48 in the group of [2] as `Logarithmic_Critical_Endpoint_Lambert_Charts/`.

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